FarmGPU

Storage requirements for neoclouds

FarmGPU Overview

On-Demand GPU compute cloud:
Spin up container / VM in seconds with latest GPUs

Cost effective:
Up to 90% less expensive than tier 1 CSP

Vendor	H100 $/hr
FarmGPU	$3
AWS	$7.5

Reliability and Security: AI ready data center, 99.9% availability, ISO/IEC security standards, confidential computing

FarmGPU Differentiation and IP

Telemetry and analytics on AI workloads
Comptute, network, and storage traces and workload analysis

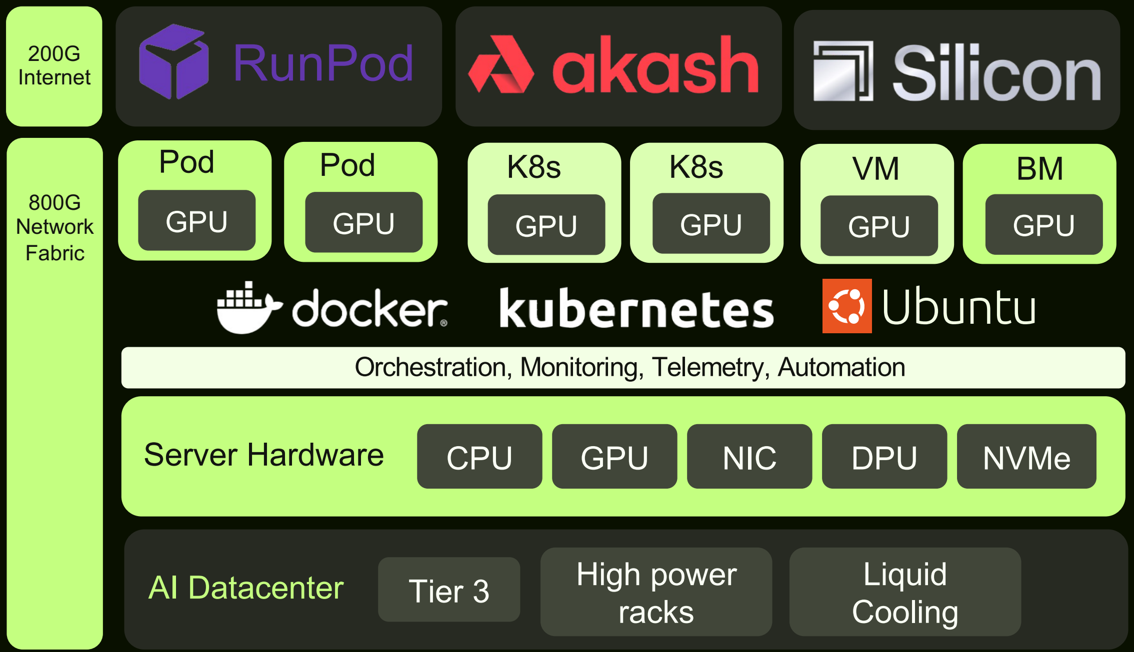
Storage Expertise
PCIe 5.0 NVMe, NVMe-oF, DPUs, storage partner ecosystem

Own High-Demand GPUs, Earn Sustainable Returns:
Enabling a new asset class for GPU ownership and strong, passive income on blockchain

FarmGPU Cloud Platform

FarmGPU

- AI Data Center Blueprint: Transforms existing data centers into optimized AI facilities (45-130kW racks, 800G networking, liquid & immersion cooling).
- On-Demand Compute Platform: Instant access to powerful NVIDIA GPUs – 70% lower cost than hyperscalers – on RunPod Secure Cloud.
- Storage for AI: Accelerates training with GPU-direct storage & BlueField DPUs; enables scalable inference via vectorDB integration.



NVIDIA direction - Fast Sparse Access to Massive Data

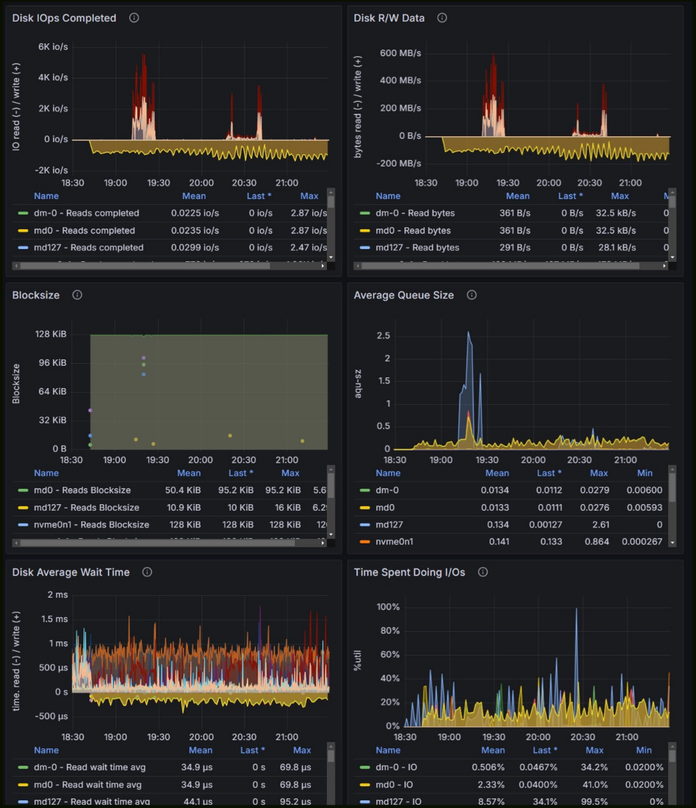
- RAG / Vector DB : 1's of TB
- Data Analytics: 10s of TB
- Graph neural networks and Relational DBs: up to 100TB
- Vector search (hyperscale): Tens of PB

GPU initiated storage workloads, memory bottlenecks, TCO benefit of flash.

Storage labs need access to GPUs, DPUs, and high-speed switches!

FarmGPU Storage Telemetry

- Custom metrics and monitoring stack
- Identify storage intensive workloads and bottlenecks
- Tracking all NVMe SMART and custom log pages
- Kernel stats
- Storage workload characterization: IOPS, bandwidth, queue depth, blocksize, iowait



Storage needs for neocloud - table stakes

Protocols: file (NFS, SMB), parallel file / pNFS, object (S3), NVMe-oF block

Scalability and performance: tiering, high performance and low node count, linear scalability

Data protection: EC, RAID, multi node, snapshot, replication, cloning, caching, quotas

Data reduction: compression, deduplication

Security: encryption, data at rest and in flight, compliance, media sanitization

Management: REST API, dashboard

Observability: telemetry, monitoring, and real time dashboard (compute, memory, storage, network, NVMe SMART)

Sustainability: storage power consumption, LCA of components

Storage needs for neocloud - best in class

DPU support, GPU direct storage, offloads for encryption, compression, virtualization, network / RDMA

Support for vector database, RAG, and SSD offload

High performance object storage with S3 endpoint

High bandwidth parallel file system for checkpointing

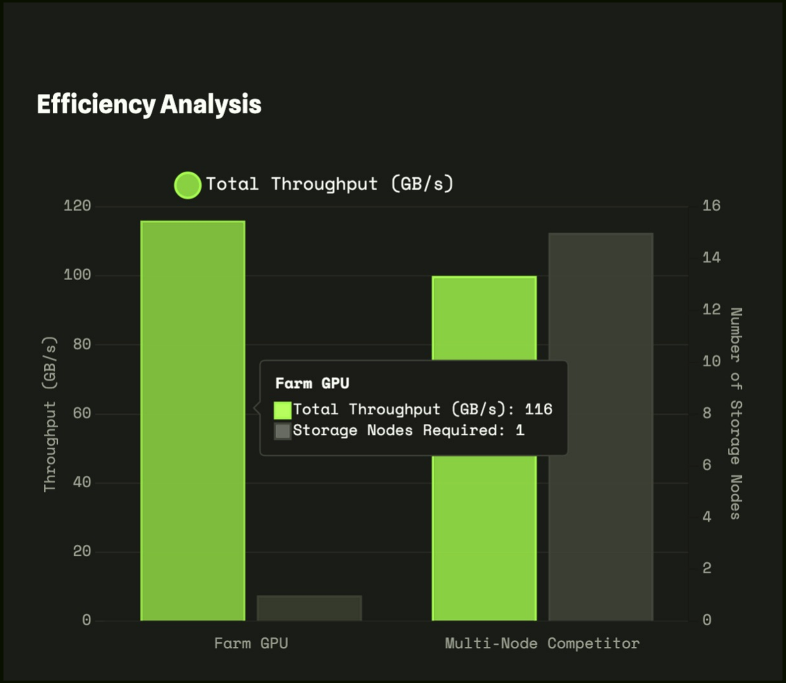
KV Cache support, NVIDIA dynamo, vLLM

Security: confidential computing, hardware root of trust

Manageability: full inventory DCIM integrations

Sustainability: runtime power optimizations, Scope 1/2/3, circularity and reusability

Fastest Single Client and Single Socket Storage Server



ML Perf 2.0 Storage Results

FarmGPU

Single Host Node Configuration

- Server Model: Supermicro SYS-212H-TN
- Form Factor: 2U Rack Mount
- Motherboard: X14SBH
- CPU: 1 x Intel® Xeon® 6781P, 80 cores, 160 threads (enabled by Hyper-Threading), 2.0 GHz base frequency, 3.8 GHz max turbo frequency, 3.2 GHz all-core turbo frequency, 336 MB L3 Cache, 350 W TDP, FCLGA4710 Socket.
- Memory: 256GB (8x32GB) Micron Technology MTC20F2085S1RC64BD2 UXCC 6400 MT/s ECC DDR5 RDIMM.
- Storage: 24 x Solidigm D7-PS1010 (SB5PH27X076T) 7.68TB PCIe 5.0 NVMe SSDs in a RAID-0 configuration
- Networking: 1 x NVIDIA ConnectX-7 MCX75310AAS-NEA Ax 400GbE / NDR IB (default mode) Single-port OSFP PCIe 5.0 x16 NIC

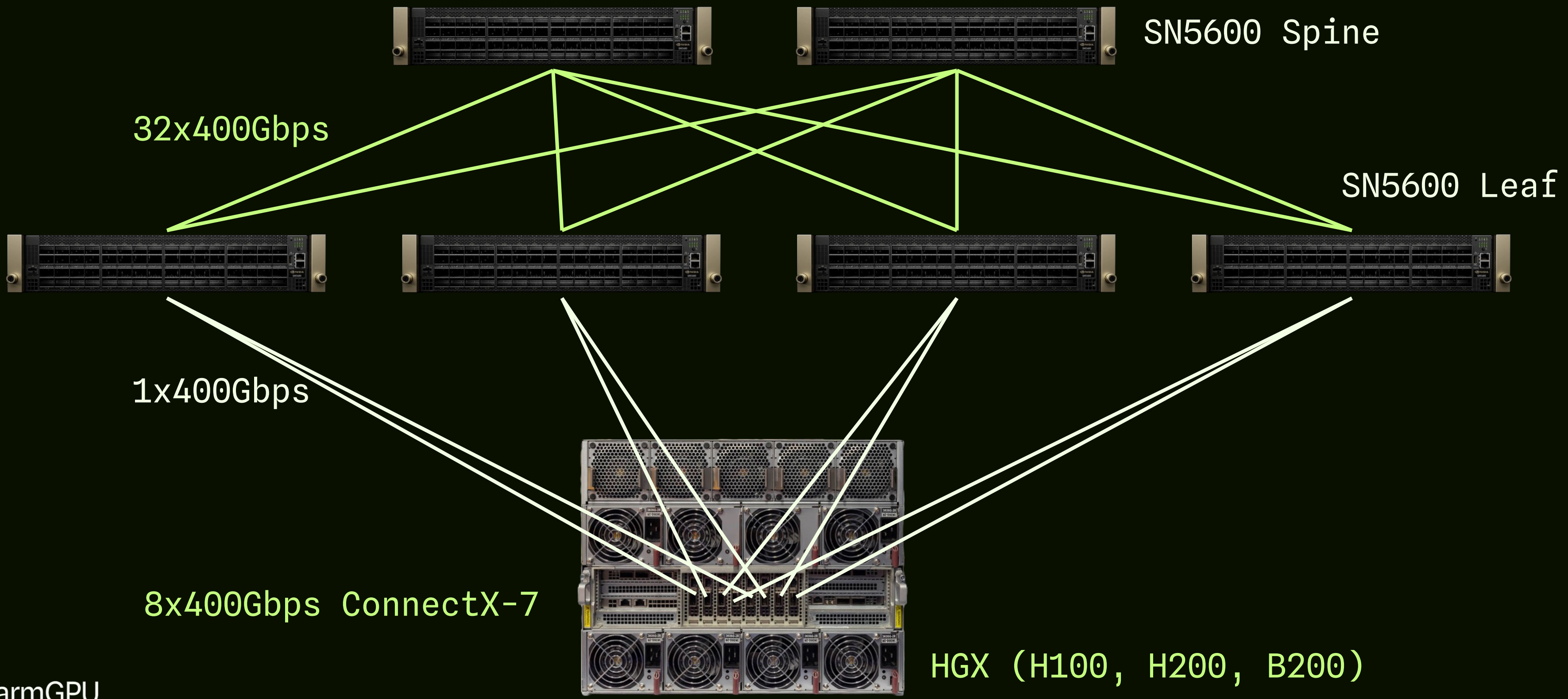
ML Perf - translation to business value

Tweaks to ML Perf storage training in PyTorch translated into 57% reduction in training time

Metric	Without Our Optimizations	With Our Optimizations	Business Implication
Time to Complete 3D U-Net	66.92 minutes	28.75 minutes	57% Faster Time-to-Insight
Total Energy Consumed	0.9018 kWh	0.3891 kWh	57% Lower Operational Costs

NVIDIA Reference Design 32 Node

Spectrum-X Ethernet E/W Compute Fabric



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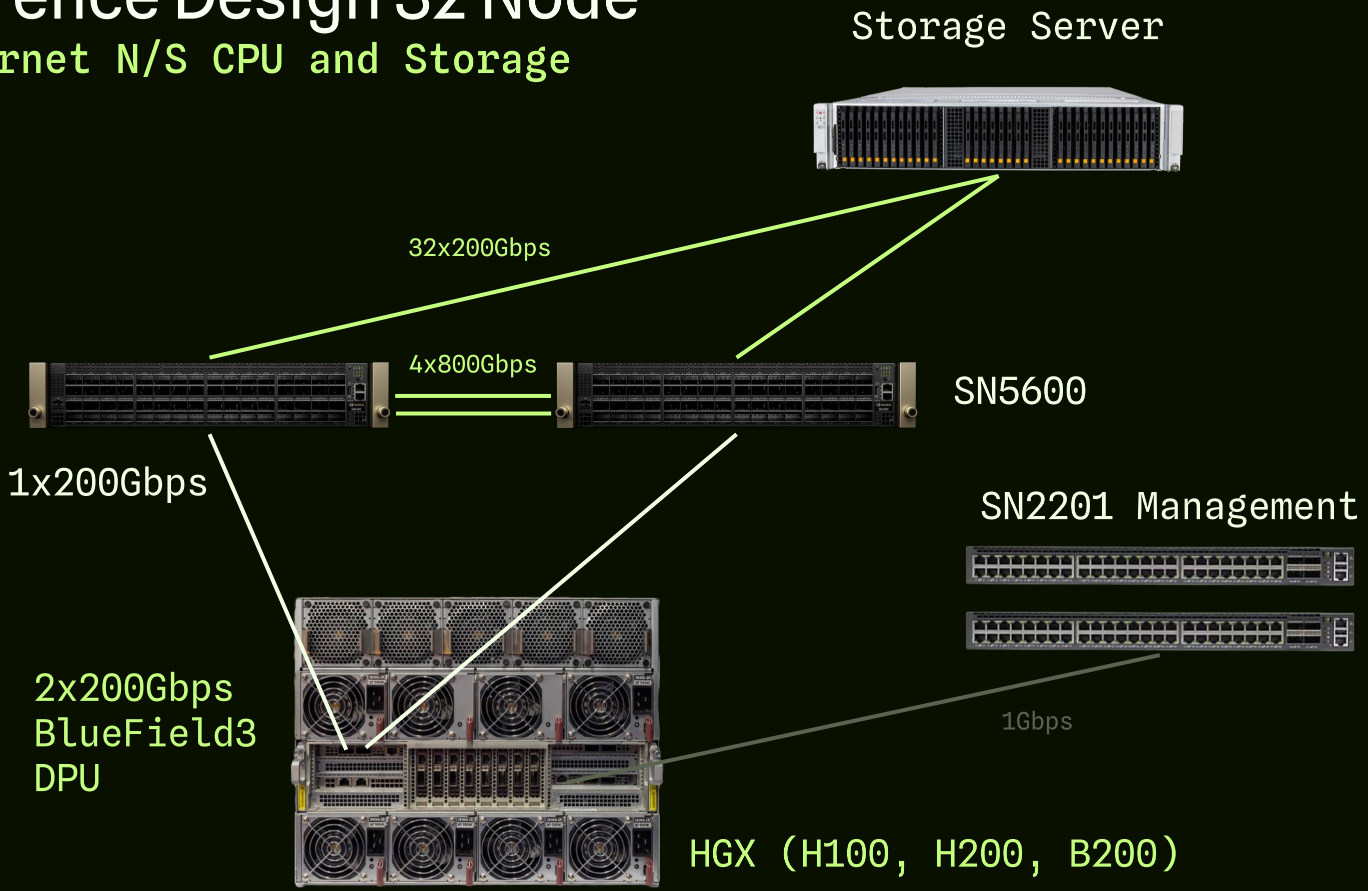
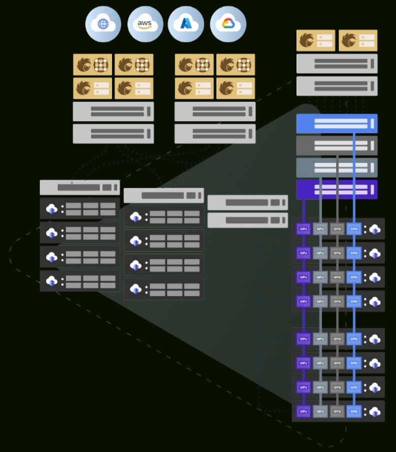
NVIDIA Reference Design 32 Node

Spectrum-X Ethernet N/S CPU and Storage



Our network design - hyperscale

- 800G Celestica DS5000
- 50% lower cost than NVIDIA reference design
- OCP Design, open specs
- SoNIC NOS
- Higher bandwidth per storage node (800Gbps vs 400Gbps)
- HedgeHog cloud open network fabric and control plane
- Custom optics tuned for CX7 and BF3 support
- Support RoCEv2 and NVMe-oF



Summary

- Neocloud is different from hyperscale and enterprise requirements
- Lots of room for optimization on AI workloads, we demonstrated almost 3x speedup from unoptimized baseline
- Lots of options for networking, but for neocloud lowest TCO wins